AXEL HILS

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https://github.com/shelovesmoussey/generic-work

PROFILE

US and French Citizen with a background in Mathematical Physics. Specialized in applying rigorous physical-mathematical methods and stochastic modeling to complex systems. Ranked 4th in an international Master's program with a focus on high-level theoretical physics and computational modeling.

EDUCATION

Université de Strasbourg Strasbourg, France
Master's Degree (M2) in Pure Mathematics (MF) 2026 – 2027
- **Ranking: TBA**
- **Relevant Coursework:** Stochastic calculus, statistical mechanics, optimal transport theory and probability, stochastic models applied to blockchain-type systems and population dynamics.

Université de Bourgogne Europe Dijon, France
Master's Degree (M2) in Mathematical Physics (Math4Phys) 2025 – 2027
- **Ranking: 4 / 25**
- **Relevant Coursework:** Stochastic processes, random matrix theory and physics, random walk on graphs, Riemann surfaces, algebraic geometry, Lie groups and algebras, quantum field theory, general relativity, path approach in QFT, Hilbert schemes and moduli spaces.

Master's Degree (M1) in Mathematical Physics (Math4Phys) 2024 – 2025
- **Ranking: 4 / 15**
- **Relevant Coursework:** Stochastic methods, Partial Differential Equations (PDEs), differential geometry, functional analysis, classical mechanics, quantum mechanics.

Bachelor's Degree (L3) in Pure Mathematics 2023 – 2024
- **Ranking: 6 / 50**
- **Relevant Coursework:** Differential calculus, topology, algebra, functional analysis, probability.

Lycée Carnot Dijon Dijon, France
Preparatory Class for Grandes Écoles (CPGE) - PCSI/PC 2021 – 2023
- Two-year intensive program in Mathematics, Physics, and Chemistry preparing for national competitive exams.

EXPERIENCE

Mathematics Teaching (Private Lessons and Tutoring) 2023 – Present
- Mathematics lessons at preparatory-class (CPGE) level and beyond (university level) for over three years, following earlier peer tutoring.
- Supporting students in understanding concepts, exercise methodology, and competitive exam preparation.

SELECTED RESEARCH

Graduate Research Thesis: Lévy Processes in Financial Modeling 2025 – 2026
- Developed mathematical frameworks using Lévy processes to model non-Gaussian market discontinuities and "jump" risks.
- Investigated infinitesimal generators and stochastic differential equations for asset price modeling.

Research Project: Binary Black Hole Solutions via the Ernst Equation 2024 – 2025
- Modeled gravitational dynamics using the Ernst equation within General Relativity.
- Applied numerical methods and complex analysis to identify stable physical solutions for binary systems.

SKILLS

- **Programming:** Python (NumPy, SciPy), C++, SQL, LaTeX, MATLAB (5+ Years).
- **Mathematics:** Stochastic Methods, Functional Analysis, Differential Geometry, Lie Algebras.
- **Physics:** Quantum Field Theory, General Relativity, Classical & Quantum Mechanics.
- **Languages:** English (Native/Bilingual), French (Native/Bilingual), Spanish.

LEISURE & DISTINCTIONS

- **Chess:** Elo 1800 – Advanced pattern recognition and strategic optimization.
- **Interests:** Clash Royale : #235 French player, Weightlifting, Cinema, and thorough reading of scientific literature.